Lab Exercise 4 : Implementation of the Sieve of Eratosthenes

Objective

To implement the **Sieve of Eratosthenes algorithm** in Java for generating all prime numbers up to a given number **n** (inclusive).
This exercise will strengthen your understanding of algorithm design, arrays, and control structures in Java.

Aim

To write a Java program that computes all prime numbers up to a given limit using the **Sieve of Eratosthenes** technique.

Theory

The **Sieve of Eratosthenes** is an ancient algorithm for finding all prime numbers up to a given limit **n**. It works by iteratively marking the multiples of each prime number starting from 2, and the numbers that remain unmarked are primes.

Algorithm Steps:

1. Create a boolean array isPrime[0...n] and initialize all entries as true.
2. Mark isPrime[0] and isPrime[1] as false since 0 and 1 are not prime.
3. For every number p starting from 2 up to √n:
    - If isPrime[p] is true, mark all multiples of p (from p*p to n) as false.
4. Finally, all indices marked true in the array correspond to prime numbers.

Example:
Input: n = 10
Output: 2 3 5 7

Procedure

1. Open your Java IDE or text editor.
2. Create two files named:
    - Sieve.java
    - Main.java
3. Implement the following code skeleton and complete the missing logic in Sieve.java.

Code Skeleton

File: Sieve.java

```java
import java.util.*;

public class Sieve {
   /**
    * TODO: Implement the Sieve of Eratosthenes algorithm.
    * This method should return a list of all prime numbers <= n.
    *
    * @param n the upper limit
    * @return list of primes up to n
    */
   public static List<Integer> sieveOfEratosthenes(int n) {
      // TODO: implement
      return new ArrayList<>();
   }
}
```

File: Main.java

```java
import java.util.*;

public class Main {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int n = sc.nextInt();
      List<Integer> primes = Sieve.sieveOfEratosthenes(n);
      for (int p : primes) {
         System.out.print(p + " ");
      }
      sc.close();
   }
}
```

Expected Output

Input:

10

Output:

2 3 5 7

Evaluation Criteria (Total Marks: 10)

Test	Description	Marks
Program compiles successfully	No syntax or compilation errors	2
Correct implementation of sieve algorithm	Generates correct primes	3
Proper use of loops and arrays/lists	Efficient and logically sound code	2
Output format correct	Matches expected format	1
Code readability and comments	Proper indentation and comments	2
Total		**10**

Result

The program successfully implements the Sieve of Eratosthenes algorithm and generates all prime numbers up to a given number **n**.